

Cellular Biophysics (BIO516) – Quiz 3

Marks: 20, Time: 1 hr

Date: 15/04/2026

Instructions to candidates

1. This paper consists of four sections: Section A (Multiple Choice: Q1-Q9), Section B (Fill in the Blanks: Q10-Q15), and Section C (Numerical Problems: Q16-Q17), Section D (subjective: Q18)

2. Answer all the questions

3. Use of calculator is permitted.

4. Standard Constants:

- Gas Constant (R) = $8.315 \text{ J}/(\text{mol} \cdot \text{K})$
- Faraday's Constant (F) = $96,485 \text{ C/mol}$ (or $96,485 \text{ J V}^{-1} \text{ mol}^{-1}$)
- Temperature (T) = 298 K (unless specified)

Section A

[Marks: 9 X 1]

Q1 During an action potential, why doesn't the membrane potential ever reach the true Nernst potential for sodium (+62 mV)?

- A) Sodium channels are permanently destroyed after opening.
- B) The rising potassium conductance limits and opposes the depolarization.**
- C) The Na/K pump instantly removes all sodium from the cell.
- A) Extracellular chloride floods into the cell to neutralize the charge.

Q2. A voltage-gated sodium channel is currently in the "inactivated" state. Conceptually, what must happen before this channel can return to its resting, closed state to be ready for the next signal?

- A) **The cell membrane must repolarize.**
- B) ATP must bind to the channel
- C) The channel must undergo further depolarization.
- D) Calcium must block the pore.

Q3. How does a single epinephrine molecule conceptually lead to the mobilization of thousands of glucose molecules?

- A) By acting as a widespread transcription factor in the nucleus.
- B) By physically carrying glucose out of the cell
- C) Through a catalytic amplification cascade where each step activates multiple downstream molecules.**
- D) By permanently fusing multiple GPCRs together.

Q4. How does β -arrestin conceptually cause "desensitization" even when a hormone is still bound to the receptor?

- A) It physically blocks the G-protein from interacting with the receptor.**
- B) It destroys the hormone in the extracellular fluid.
- C) It rapidly degrades adenylyl cyclase.
- D) It pumps all cyclic AMP out of the cell.

Q5 Which of the following is the "Odd One Out" regarding the termination of a G-protein coupled signalling response?

- A) Intrinsic GTPase activity of the G_{sa} subunit.

- B) Phosphodiesterase-mediated hydrolysis of cAMP to 5'-AMP.
- C) Phosphorylation of the β -adrenergic receptor by BARK/GRK2.**
- D) Dephosphorylation of target proteins by intracellular phosphatases.

Q6. Consider the following statements regarding the Sodium Channel model:

- (i) Transition from I_2 to C_3 is reversible.
- (ii) The channel can inactivate without ever opening.
- (iii) In State I_1 , all activation gates are closed but the pore is blocked.
- (iv) The I_1 state is thermodynamically favoured during maintained depolarization.

Which combination is not incorrect?

- A) (i), (ii), and (iii)
- B) (i), (ii), and (iv)**
- C) (ii) and (iv) only
- D) All are correct.

Q7 In the Hodgkin-Huxley model, "Delayed Rectifiers" refer to:

- A) Sodium channels that inactivate slowly.
- B) Voltage-gated Potassium channels because they peak after sodium conductance declines.**
- C) The leak conductance path for Chloride.
- D) The inactivation gate of the sodium channel.

Q8 A is an agonist of compound Z while P is a toxic structural analog of Z that can compete for the same binding environment but lacks intrinsic activity. E is the antagonist of compound W while K is the structural and functional analog of W. Z and W are known to interact with each other and make response out of the receptor 9YBA, a hypothetical receptor which further activates the division of stem cells into blood cells. After the cells is taken out from one of the subjects, one of my collaborator is considering to use the combination of A and E to stimulate the natural path followed by the cell to continue with development of blood cells out of stem cells. What are they doing wrong?

- A) They are completely right they should keep using A and E to stimulate the development.
- B) They are partially correct, they should be using A only to activate the development.
- C) They are partially correct, they should use E only as P is a toxic compound.
- D) The approach fails because antagonism of W prevents formation of the functional signaling complex required for receptor activation, and a partial agonist like K would be more appropriate.**

Q9 For the previous question, evaluate the following statements:

- (i) P can reduce the efficacy of A despite lacking intrinsic activity.
- (ii) A and K together can produce submaximal but sufficient activation of receptor 9YBA to induce differentiation.
- (iii) Complete activation of receptor 9YBA strictly not necessarily interaction between Z and W.
- (iv) Only one compound is sufficient to cause the activation of 9YBA receptor.
- (v) Receptor activation is determined not only by ligand binding but also by cooperative interactions between signaling partners.

Which of the following combinations is not incorrect?

- A) iii, and iv only
- B) ii and iv only**

- C) i, ii, and v only
D) i, ii, iii, and v only

Section B

[Marks: 6 X 1]

Q10. The action potential relies on two contrasting mechanisms: a _____ feedback loop driven by sodium channels that causes the rapid upstroke, and a self-limiting _____ feedback loop driven by delayed rectifier potassium channels that opposes depolarization. (Ans: **Positive** and **negative**, respectively)

Q11. The process of removing receptor molecules from the plasma membrane via endocytosis to dampen a continuous signal is called receptor _____. (Ans: **sequestration**)

Q12 In the cyclic AMP system, the drug _____ prolongs the hormonal response by inhibiting the enzyme phosphodiesterase. (Ans: **theophylline**)

Q13. In the sodium channel model, the transition between State I₁ and State I₂ implies that the blocking segment can prevent conduction while allowing a _____ to close and reopen. (Ans: **single gate**)

Q14. The protein _____ binds to the phosphorylated carboxyl-terminus of the GPCR to prevent further interaction with G-proteins. (Ans: **β-arrestin (or Barr)**)

Q15. _____ can act as a tumor promoter but constantly activating PKC. (Ans: **Myristoylphorbol acetate/ Phorbol esters**)

Section C

Q16. Using the complete electrical analog model of the axolemma (which includes the leak pathway), the resting membrane potential of a specific neuron is exactly -90 mV. Assume the Nernst potentials are $E_{Na} = +62$ mV, $E_K = -95$ mV, and the leak potential is $E_l = -87$ mV. If the resting potassium conductance (G_K) is 0.50 mS/cm² and the leak conductance (G_l) is 0.30 mS/cm², calculate the exact resting sodium conductance (G_{Na}) in mS/cm². Round your answer to four decimal places.

[Marks: 2]

Use equation 9.6 in class 18 note, rearrange the equation.

Ans:

$$G_{Na} = -\frac{G_K(V_m - E_K) + G_l(V_m - E_l)}{V_m - E_{Na}}$$

First, compute the driving forces:

$$\begin{aligned} V_m - E_{Na} &= -90 - 62 = -152 \text{ mV}, \\ V_m - E_K &= -90 - (-95) = 5 \text{ mV}, \\ V_m - E_l &= -90 - (-87) = -3 \text{ mV}. \end{aligned}$$

Now substitute into the equation:

$$G_K(V_m - E_K) = 0.50 \times 5 = 2.5,$$

$$G_l(V_m - E_l) = 0.30 \times (-3) = -0.9.$$

Sum of potassium and leak currents:

$$G_K(V_m - E_K) + G_l(V_m - E_l) = 2.5 + (-0.9) = 1.6$$

Thus,

$$G_{Na} = -\frac{1.6}{-152} = \frac{1.6}{152} = 0.01052631578947 \dots$$

Rounded to four decimal places:

$$\boxed{0.0105}$$

The resting sodium conductance is $G_{Na} = 0.0105 \text{ mS/cm}^2$.

Q17. In a mammalian axon, the axoplasmic and extracellular concentrations of Cl^- ion are 4 mM and 103 mM, respectively. Determine the magnitude and direction of the driving force for Cl^- flux, (in mV) for the passive transport of the ion across the cell membrane when the cell potential is -90 mV (inside negative) and $T = 37^\circ\text{C}$. [Marks: 2]

Ans: Driving force calculation for **outside to inside transport**

Driving force due to chemical gradient across membrane = 86.7 mV

Driving force due to electrical gradient across membrane = -90 mV

Driving force due to electrochemical gradient across membrane = **-3.3 mV**

So, passive diffusion will be from inside to outside with a driving force 3.3 mV

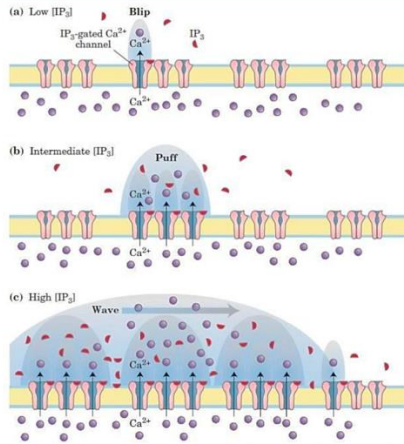
Section D

Q18. Differentiate between blips, puffs and waves.

[Mark: 1]

Ans:

Occurrence of localized Ca^{2+} “blips”, “puffs”, and “waves”



The activity of the second messenger Ca^{2+} , can be spatially restricted; after its release triggers a local response, Ca^{2+} is generally removed before it can diffuse to distant parts of the cell. Very commonly, Ca^{2+} level does not simply rise and then decrease, but rather oscillates with a period of a few seconds even when the extracellular concentration of the triggering hormone remains constant. The mechanism underlying $[\text{Ca}^{2+}]$ oscillations presumably entails feedback regulation by Ca^{2+} on some part of the Ca^{2+} -release process. Whatever the mechanism, the effect is that one kind of signal (hormone concentration, for example) is converted into another (frequency and amplitude of intracellular $[\text{Ca}^{2+}]$ “spikes”). Another variation is the occurrence of localized Ca^{2+} “blips,” “puffs,” and “waves”—transient increases in $[\text{Ca}^{2+}]$ that are limited to specific subcellular regions. The Ca^{2+} signal diminishes as Ca^{2+} diffuses away from the initial source (the Ca^{2+} channel), is sequestered in the ER, or is pumped out of the cell.